

AI-Powered Astronaut Health Monitoring System

NASA Space Mission AI Project | ITAI 2372

STUDENT

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Group 01

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Conceptual Design Track

Presentation Overview

- Problem Statement — Why astronaut health monitoring needs AI
- Proposed Solution — The AHMAI system overview
- AI Architecture — Four-layer ensemble model design
- Data Pipeline — What data the system uses and how it's processed
- Alert System — How crew and mission control are notified
- Testing Plan — How the system will be validated
- Ethical Considerations — Bias, privacy, and safety
- Conclusion & References

Problem Statement

Why does space health monitoring need AI?

Health Risks in Space

- Cardiovascular deconditioning
- Immune system suppression
- Cognitive degradation
- Bone density loss (1-2%/month)
- Radiation exposure

Current Monitoring Gaps

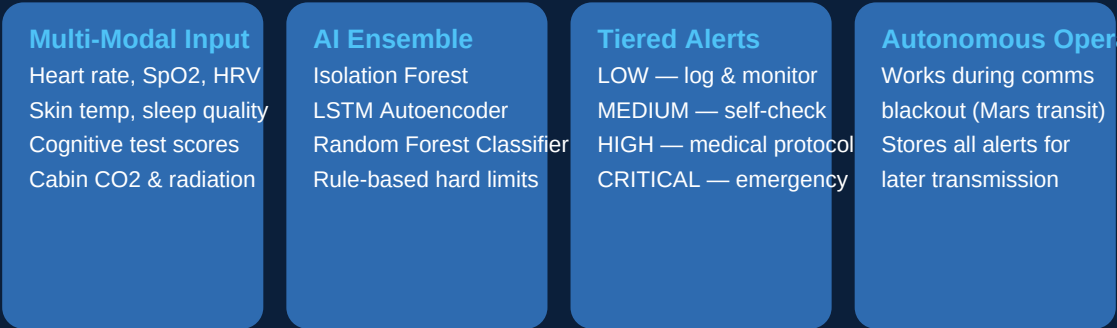
- Mars signal delay: 3-22 minutes
- Manual review can't catch subtle multi-day patterns
- No real-time autonomous alerts
- Fixed thresholds miss individual variation in trained astronauts

An autonomous AI system is needed that works even when Earth is unreachable.

Proposed Solution

Astronaut Health Monitoring AI (AHMAI)

A multi-layer AI ensemble that continuously monitors biometric and environmental data, detects anomalies against each astronaut's personal baseline, and alerts crew in real time.



AI Architecture — Four-Layer Ensemble

LAYER 1: Rule-Based Hard Limits

Always active. Instant CRITICAL alert if SpO2 < 90%, HR > 160 bpm at rest, or cabin CO2 > 5,000 ppm.
No ML delay.

LAYER 2: Isolation Forest

Unsupervised anomaly detection on 5-min sensor windows. No labeled data needed. Flags score > 0.72.

LAYER 3: LSTM Autoencoder

Learns normal time-series rhythms. High reconstruction error = anomaly. Retrained monthly on mission data.

LAYER 4: Random Forest Classifier

Classifies flagged anomalies into LOW / MEDIUM / HIGH / CRITICAL and explains top contributing features.

Data Pipeline

What the system monitors and how it processes it

Sensor Inputs

Heart Rate	1 Hz — ECG wearable patch
SpO2	0.2 Hz — pulse oximeter
Heart Rate Variab.	Derived — autonomic indicator
Skin Temperature	0.1 Hz — circulation proxy
Activity Level	50 Hz — accelerometer
Sleep Quality	Nightly — EEG headband
Cognitive Tests	Daily — reaction time, memory
Cabin CO2	0.017 Hz — environment sensor
Radiation Dose	1 Hz — dosimeter

Preprocessing Steps

- 1 Artifact removal (z-score > 3.5 SD flagged)
- 2 Personalized normalization vs. 14-day baseline
- 3 Window aggregation: 5-min / 1-hr / 24-hr
- 4 Feature engineering (12 derived features)

Alert & Output System

How the system communicates findings

CRITICAL

Audio alarm + wearable vibration + screen

Latency: < 5 seconds | Notifies: Crew immediately

HIGH

Screen notification + mission control feed

Latency: < 60 seconds | Notifies: Crew + Earth

MEDIUM

Screen notification at next crew check-in

Latency: < 5 minutes | Notifies: Crew

LOW

Logged to daily 24-hour health summary

Latency: Daily report | Notifies: Crew (daily)

Example: "SpO2 declining for 6 hrs (94% vs 98% baseline). Pattern matches early respiratory fatigue. Recommended action: perform lung capacity self-test from medical kit."

Testing Plan & Ethical Considerations

Testing — 4 Phases

- Phase 1 — Data validation: artifact removal, gap handling, normalization
- Phase 2 — Model performance: synthetic anomaly injection, precision ≥ 0.85 , recall ≥ 0.90
- Phase 3 — Integration: alert latency, graceful degradation, comms blackout simulation
- Phase 4 — Bias audit: demographic subgroup analysis, explainability review

Key Metrics

Precision	≥ 0.85
Recall	≥ 0.90
F1 Score	≥ 0.87
ROC-AUC	≥ 0.92
CRITICAL latency	$< 5 \text{ sec}$
False positive rate	$< 5\%$

Ethical Considerations

- Privacy — data stored locally, transmitted only during scheduled downlinks
- Bias — personalized baselines reduce dependence on population averages
- Over-reliance — AI supports crew judgment, does not replace it
- Fail-safe — system reverts to rule-based layer if ML models go offline

Key Takeaways

- Long-duration spaceflight creates health risks that current monitoring cannot catch in time
- The AHMAI four-layer ensemble detects anomalies faster and more accurately than rule-based systems
- Personalized baselines eliminate population-average bias for individual astronauts
- The system operates fully autonomously during Mars-transit communication blackouts
- Ethical design ensures crew trust, privacy, and safe failure modes

References

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